

GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

QIAN QI

ABSTRACT. The geometry obtained by a continuing experiment is weighted by the time for which its predictions are retained, not merely by the law used to prepare it. We prove a causal transfer theorem for regenerative experiments operated by one parameter-independent finite-state controller. The joint physical preparation and observer reset is an explicit, charged protocol. Finite-history product domination and a controller-uniform return-tail envelope yield an attained robust average risk, determination by finite model families, and uniform comparison with finite-acquisition and vanishing-discount values at the unchanged state budget. A return moment of order $1 + p$ gives the sharp uniform orders N^{-p} and $(1 - \beta)^p$. A second sharp transfer exponent, $p/(1 + p)$, converts a scored-cycle causal defect into average-risk error. Primitive kernel and instrument errors give effective lower certificates against every same-budget Borel controller and upper certificates implemented by finite tables. On the actual renewal-occupation law, small-ball mass, global covering and occupation-weighted predictive stability give matching checkpoint and online memory curves. Hidden-state experiments and noncommuting quantum instruments verify the assumptions independently; heavy-tailed and rank-changing classes are included. The theorem does not identify a compulsory observer reset with physical mixing, and does not infer an unrestricted average-cost theorem by formally removing a discount.

1. INTRODUCTION

A prepared causal experiment determines which future tests can actually be performed. Their conditional laws determine predictive equivalence. An observer, however, must realize those predictions through one executable program, and the histories on which that program spends its resources are distributed according to the experiment it really runs. These three restrictions—causal realization, actual acquisition, and finite resources—must be kept together when transferring a geometric risk bound from one experiment to another.

There are two obstacles to a long-run theory. First, a finite controller reuses the same report row at every time. A finite-history static reduction does not by itself establish continuity under this reuse. Second, neither bounded mean return times nor a separate result at every discount controls the long-run optimization uniformly. A rare preparation can occupy a substantial fraction of physical time. Its statistical mass and its resource cost must not be removed by conditioning.

We resolve these obstacles for a class defined at the experiment level. A cycle consists of a charged preparation and a causal experiment ending at a visible regeneration boundary. At that boundary the physical preparation is renewed and the observer is forcibly returned to its distinguished state. One stationary M -state program is used in every cycle and at every inaccessible model parameter. The reset is part of the protocol, not a consequence attributed to a mixing transition kernel. The within-cycle experiment may be nonlinear, nonmixing, infinite-dimensional, history-dependent, or have changing report supports. The exact theorem assumes domination

Date: 9 October 2026.

2020 Mathematics Subject Classification. Primary 62B05, 93E20; Secondary 60K05, 60G35, 94A34, 81P45.

Key words and phrases. causal experiment, predictive quotient, renewal occupation, finite-state controller, average risk, vanishing discount, acquired geometry, quantum instrument.

of each finite report word by a common product reference; it does not assume domination of an infinite path law.

The central result is Theorem 2.2. Its hypotheses involve a uniform tail envelope for actual cycle lengths, before a program is optimized. Its conclusions compare the same program class under three resource interfaces: N real acquisition calls, a normalized discount, and average risk. Under a return moment $\sup_{\theta, \gamma} \mathbb{E}_\theta^\gamma \tau^{1+p} \leq K$, $0 < p \leq 1$, they imply

$$(1) \quad |V_N^\Theta(M) - V_{\text{av}}^\Theta(M)| \leq 2KN^{-p}, \quad |V_\beta^\Theta(M) - V_{\text{av}}^\Theta(M)| \leq 2K(1 - \beta)^p.$$

No memory is added when the acquisition horizon or discount changes. The bounds concern expected risk uniformly over models and programs; they are not uniform sample-path deviation bounds. Proposition 3.3 shows that the orders in (1) cannot be improved uniformly over the stated moment class.

The acquired geometry is the cycle occupation measure divided by the actual mean cycle length. Theorem 5.1 bounds online prediction between quantization of that measure and an implementable recursive covering estimate. It uses global covers and stability along actual acquisitions, including all candidate states used by the recursion. Small-ball mass then gives matching curves. Theorems 6.3 and 6.5 obtain their dimension and acquired mass from raw likelihoods and positive instruments, not from a supplied covariance or Fisher matrix. The stability criterion is an occupation-weighted one; a separate heavy-tailed holding class has an exact, possibly singular, memory law without filter contraction.

The resource composition is also sensitive to recurrence. A small total-variation defect on one completed scored cycle need not produce a linear error in time-average risk. Theorem 4.1 gives a tail-dependent modulus, and a $1 + p$ moment yields order $\varepsilon^{p/(1+p)}$. Proposition 7.1 realizes this exponent together with a matched memory term and an inaccessible calibration coordinate in one task. Primitive kernel and unnormalized instrument bounds are translated into all-controller moduli in Lemma 4.2. Thus the effective Theorem 4.4 does not leave its decisive finite-response error unspecified. Its enumeration is finite, not efficient; table description, report parsing, scratch, and planning cost remain distinct from M .

Renewal-reward identities, overshoot estimates, compactness, and finite-intersection arguments are classical tools [1, 9, 6]. The finite-row tensor argument used here is the one isolated in the discounted common-controller analysis [11]. The new synthesis is the uniform renewal passage at fixed observer size, its sharp defect and time moduli, and its coupling to actual occupation geometry and primitive effective certificates. Section 8 compares this result with strategic-measure existence, average-cost POMDP theory, and finite-memory approximation. In particular, we do not claim that the existence of an average-cost control theory itself is new.

2. PREPARED CYCLES AND THE TRANSFER THEOREM

2.1. Raw causal data and the reset protocol. Let $\Theta \neq \emptyset$ be a set. A realized θ is fixed for the entire execution. In each cycle a standard Borel hidden state is prepared, and positive normalized causal kernels produce the next hidden state, a report Y_t , an audit mark Z_t , and a visible end-of-cycle flag. Positive means nonnegative. Hidden state may contain the necessary physical history. Audit marks are not feedback to the observer. They are actual jointly specified marks of the performed kernels; in a quantum realization, any audit back-action is included in the instrument. We do not posit a joint path of unperformed incompatible audits.

There is a nonempty finite action set A at every ordinary data call, with all its actions legal. The compulsory preparation call and the end flag are prescribed interfaces, not extra decisions. A cycle has an integer length $\tau \geq 1$ counting *all* its raw calls, including preparation. The end flag makes τ a stopping time of visible within-cycle history. At the next preparation, the physical system is freshly prepared with the same θ and the observer is overwritten by state 1. No private seed,

counter, or record may survive this overwrite. Distinct cycles use independent physical innovations and fresh controller randomization, conditional on θ . Reset calls have a prescribed bounded score; the predictive examples below give them score zero, but still charge their acquisition and time costs.

Reports for action a lie in a standard Borel space Y_a . For every model, finite within-cycle intervention word $\mathbf{a} = (a_1, \dots, a_t)$, and its prescribed preparation positions, assume

$$(2) \quad P_{\theta,t}^{\mathbf{a}} \ll \sigma_{a_1} \otimes \dots \otimes \sigma_{a_t},$$

where each σ_a is one probability measure common to the family. The reports include the visible flags. A stopped prefix is padded by a fixed cemetery report if necessary, and each reference includes the required atoms. Only finite-history product domination is asserted. Moving or zero densities are allowed. One-time marginal domination is not sufficient.

Decisions lie in a compact metrizable space D , and $0 \leq \ell(d, z) \leq 1$ is measurable and continuous in d for every z . The audit law is part of the raw kernel and does not depend on a private controller seed except through declared actions; d is a readout, not an unrecorded intervention changing the next state. Decision-dependent physical audits can instead be placed in the action interface when that interface satisfies the stated finite-action assumptions. These distinctions make the cycle reward a well-defined random variable.

Definition 2.1 (One reused observer). A program $\gamma \in \Gamma_M$ consists of action rows $\alpha_i \in \Delta(A)$, Borel report rows $k_{ia} : Y_a \rightarrow \Delta([M])$, and readouts $d_i \in D$, $i \in [M] = \{1, \dots, M\}$. At an ordinary call it draws a from α_i , receives y , draws j from $k_{ia}(y)$, and issues d_j for that call's audit. It then retains j , except that the prescribed boundary overwrite returns it to 1. The same rows are used at all times, cycles, and parameters. Sampling is fresh and conditionally independent. A compulsory preparation is processed by its prescribed overwrite, not by an extra phase-indexed row. All persistent observer registers are included in $[M]$.

A Borel row need not be computable. A finite executable program will be supplied under the stronger hypotheses of Section 4. The exact program class is not enlarged to history-dependent or phase-dependent policies when a value is compared.

Write $R = \sum_{t=1}^{\tau} \ell_t$ for one cycle's reward, where $\ell_t = \ell(d_{I_t}, Z_t)$ at data calls and is the prescribed score at preparation. Thus $0 \leq R \leq \tau$. Put

$$(3) \quad \mu_{\theta}(\gamma) = \mathbb{E}_{\theta}^{\gamma} \tau, \quad r_{\theta}(\gamma) = \mathbb{E}_{\theta}^{\gamma} R, \quad g_{\theta}(\gamma) = r_{\theta}(\gamma) / \mu_{\theta}(\gamma).$$

The actual renewal occupation measure of any within-cycle measurable state S_t is

$$(4) \quad \bar{\nu}_{\theta}^{\gamma}(B) = \frac{1}{\mu_{\theta}(\gamma)} \mathbb{E}_{\theta}^{\gamma} \sum_{t=1}^{\tau} 1_{\{S_t \in B\}}.$$

It is a probability measure if every raw call is represented, and a subprobability measure when only scored data calls are retained. It is not generally the preparation law.

2.2. Return envelope and common values. Assume there is a decreasing function $a : \mathbb{N}_0 \rightarrow [0, \infty)$ with

$$(5) \quad a(0) < \infty, \quad a(T) \longrightarrow 0, \quad \sup_{\theta \in \Theta, \gamma \in \Gamma_M} \mathbb{E}_{\theta}^{\gamma} (\tau - T)_+ \leq a(T).$$

This is a certificate for the *actual* controlled cycle law. A bounded mean alone does not imply it. It is imposed on all legal programs before optimization, including the finite tables later used for approximation.

Let $C_N = \sum_{t=1}^N \ell_t$ along the concatenated cycles, starting at a preparation. Define

$$(6) \quad \begin{aligned} J_{\theta,N}(\gamma) &= \mathbb{E}_\theta^\gamma C_N / N, & J_{\theta,\beta}(\gamma) &= (1 - \beta) \sum_{t \geq 1} \beta^{t-1} \mathbb{E}_\theta^\gamma \ell_t, \\ V_N^\Theta(M) &= \inf_{\gamma \in \Gamma_M} \sup_{\theta} J_{\theta,N}(\gamma), & V_\beta^\Theta(M) &= \inf_{\gamma \in \Gamma_M} \sup_{\theta} J_{\theta,\beta}(\gamma), \\ V_{\text{av}}^\Theta(M) &= \inf_{\gamma \in \Gamma_M} \sup_{\theta} g_\theta(\gamma). \end{aligned}$$

The discounted weights sum to one. Scores occur after the report-to-label update. The integer N counts real raw calls, including reset calls, but is an external evaluation budget: the deployed rows do not receive N or a clock. An exact internal deadline is a different protocol.

Set

$$(7) \quad b_N = \frac{1}{N} \sum_{j=0}^N a(j), \quad c_\beta = (1 - \beta)^2 \sum_{N \geq 1} N \beta^{N-1} b_N.$$

Both tend to zero in their respective limits.

Theorem 2.2 (Causal renewal transfer). *Under the prepared-cycle interface, (2), and (5), the following hold for every finite M .*

The average risk $V_{\text{av}}^\Theta(M)$ is attained by one Borel program. For each fixed model and program, $C_N/N \rightarrow g_\theta(\gamma)$ almost surely and in mean. Moreover,

$$(8) \quad V_{\text{av}}^\Theta(M) = \sup_{\emptyset \neq F \subset \Theta, |F| < \infty} V_{\text{av}}^F(M).$$

A strict target below the full value is therefore excluded by a finite subfamily of fixed models.

Uniformly in every θ and γ ,

$$(9) \quad |J_{\theta,N}(\gamma) - g_\theta(\gamma)| \leq b_N, \quad |J_{\theta,\beta}(\gamma) - g_\theta(\gamma)| \leq c_\beta.$$

The same inequalities compare the three optimized values. Every cluster point, in the controller topology described below, of asymptotically optimal finite-acquisition or vanishing-discount programs is average-optimal. Exact optimal programs at finite N and $0 < \beta < 1$ also exist.

If, in addition, for some $0 < p \leq 1$,

$$(10) \quad \sup_{\theta, \gamma} \mathbb{E}_\theta^\gamma \tau^{1+p} \leq K,$$

then b_N and c_β in (9) may be replaced by

$$(11) \quad K \frac{(N+1)^{1-p}}{N} \leq 2KN^{-p}, \quad 2^{1-p}K(1-\beta)^p,$$

respectively. Replacing a physical realization by another with the same finite report, survival and payoff responses leaves all these values unchanged, provided the same reset and resource protocol is preserved.

There is no minimax interchange in (8); the parameter is not resampled or adversarially changed at each call. The theorem gives no cardinality bound on the witnessing finite family. For $0 \leq \ell \leq H$, all risk-error bounds acquire a factor H , including a discounted prefix tail $H\beta^T$.

2.3. Predictive equivalence and its measurable realization. Fix a preparation, or a specified Bayesian preparation of an unknown parameter. Take a countable determining family of executable future tests, containing the visible interface and closed under the finite continuations needed to determine subsequent tests. Evaluation gives a Borel map

$$(12) \quad q(h) = (\mathbb{E}[f_j | h])_{j \geq 1} \in [0, 1]^\mathbb{N}.$$

The conditional versions are fixed on the actual prepared history space. No model-dependent q is provided to the robust controller as a hidden register.

Proposition 2.3 (Predictive realization). *The fibers of (12) are exactly future-test equivalence. Every sufficient statistic through which all test expectations factor also determines q . If the evaluation image is Borel, and the one-step future-test closure includes the joint next-report/test laws, the image is a standard Borel predictive realization: report laws and a Borel update descend to it, up to their actual null sets. Without the Borel-image condition, only the evaluation statistic and its stated minimality are asserted.*

Proof. Countable determining tests and the monotone-class principle identify the future laws, proving the fiber assertion. Coordinatewise factorization proves minimality. Suppose the image is Borel. A Borel map constant on the fibers of a Borel surjection between standard Borel spaces has a Borel factor: the image of a saturated Borel set and of its complement are disjoint analytic sets whose union is the target, and hence both are Borel. Apply this observation to the probabilities of a countable generating algebra of next-report/future-test events. It gives the descended joint kernels. Standard Borel disintegration then gives conditional next-test expectations as Borel functions of (q, a, y) . Their countably many versions agree with the original updated evaluations almost surely. Complete the update on the remaining Borel null fibers by a fixed image point. This proves realization without postulating a manifold or a Fisher matrix. The underlying disintegration and version facts are standard [6, 2]. $\square$

3. THE RENEWAL PASSAGE AT UNCHANGED MEMORY

Lemma 3.1 (Finite-response continuity). *Equip each simplex-valued report row with its weak-star $L^\infty(\sigma_a)$ topology, each action row with its Euclidean topology, and D^M with its compact topology. The resulting space Γ_M , modulo the common reference-null sets, is compact. For fixed θ, T , the maps*

$$\gamma \longmapsto \mathbb{E}_\theta^\gamma(\tau \wedge T), \quad \gamma \longmapsto \mathbb{E}_\theta^\gamma \sum_{t \leq \tau \wedge T} \ell_t$$

are continuous.

Proof. Positivity and the row-sum identity are weak-star closed constraints in finitely many unit balls. Banach–Alaoglu and the finite product theorem give compactness. On a standard Borel probability space, a completed-measurable representative has a Borel version. Complete all finitely many rows on one union of reference-null exceptional sets by the point mass at 1. Product domination makes this completion common to every model and finite history.

For clarity, the continuity argument applies to nets. If $u_{j,\lambda} \rightarrow u_j$ weak-star and $|u_{j,\lambda}| \leq 1$, then for any $f \in L^1(\bigotimes_{j=1}^t \sigma_j)$,

$$\int f(y) \prod_{j=1}^t u_{j,\lambda}(y_j) d\sigma(y) \longrightarrow \int f(y) \prod_{j=1}^t u_j(y_j) d\sigma(y).$$

It holds first for rectangle indicators, where the integral factors, and then for their finite linear combinations. These are L^1 -dense, and the replacement error is uniform in the net. Several factors may be copies of one changing row, because their report coordinates are distinct.

For each t , form the payoff submeasure

$$\Lambda_{\theta,t}^{\mathbf{a},d}(B) = \mathbb{E}_\theta^{\text{do}(\mathbf{a})}[1_{\{\tau \geq t, Y_{1:t} \in B\}} \ell(d, Z_t)].$$

It is dominated by the report measure in (2). Its density is L^1 -continuous in d by bounded convergence. Causal expansion gives, with $i_0 = 1$,

$$(13) \quad \mathbb{E}_\theta^\gamma[1_{\{\tau \geq t\}} \ell_t] = \sum_{\mathbf{a}, \mathbf{i}} \int f_{\theta, t}^{\mathbf{a}, d_{it}}(y) \prod_{s \in \mathcal{D}_t} \alpha_{i_{s-1}}(a_s) k_{i_{s-1}, a_s}(i_s | y_s) d\sigma_{\mathbf{a}}(y).$$

Here $\mathcal{D}_t$ consists of the ordinary data positions; forced preparation positions have their prescribed factor. The action probabilities are in the product. The expansion sums interventions; it does not condition on an event of possibly zero probability. Tensor continuity proves continuity of each finite sum. Using the survival submeasure in place of the payoff and summing over $t \leq T$ proves the first assertion too. This is the reused-row step, rather than continuity of pointwise multiplication in one L^∞ space [11]. $\square$

Remark 3.2 (Why the product hypothesis matters). For uniform $Y \in [0, 1]$, let two reports both equal Y , and let r_n be successive dyadic $\{0, 1\}$ -valued oscillations. Then $r_n \rightarrow 1/2$ weak-star, but $\mathbb{E}r_n(Y)^2 = 1/2$, not $1/4$. The report pair is diagonal and singular to product Lebesgue measure. Independent-position static reduction does not justify the missing limit. Conversely, after product domination the diagonal constraint that the same row is reused is closed in the product row topology, and Lemma 3.1 supplies its continuity. The subsequent compact minimization is standard; no new abstract compactness principle is claimed.

Proof of Theorem 2.2. The truncation errors for r_θ and μ_θ are at most $a(T)$. They are therefore continuous by Lemma 3.1 and uniform convergence. Since $\mu_\theta \geq 1$, so is g_θ . The supremum of these continuous functions is lower semicontinuous on one compact space, and attains its minimum. For a level u , the sets $\{\gamma : g_\theta(\gamma) \leq u\}$ are closed. If every finite subfamily admits a point in their intersection, compactness gives a point in the full intersection. Applying this to levels above the supremum of finite-subfamily values proves (8). The expansion proves response invariance.

For fixed θ, γ , cycles (τ_j, R_j) are iid, conditional on this fixed model. Write $T_k = \sum_{j=1}^k \tau_j$ and $K_N = \min\{k : T_k > N\}$. Since $\tau_j \geq 1$, $K_N \leq N + 1$. The centered cycle increments $X_j = R_j - g_\theta \tau_j$ have mean zero. The event $\{K_N \geq j\} = \{T_{j-1} \leq N\}$ depends only on preceding cycles; consequently

$$\mathbb{E} \sum_{j=1}^{K_N} X_j = 0.$$

Removing from the last cycle its reward and time after call N changes this centered sum by a random variable whose absolute value is at most $T_{K_N} - N$. Thus

$$(14) \quad |\mathbb{E}C_N - Ng_\theta| \leq \mathbb{E}(T_{K_N} - N) \leq \sum_{s=0}^N \mathbb{E}(\tau - s)_+.$$

For the last inequality, sum over the last renewal epoch before N . A given integer is a renewal epoch with probability at most one, because increments are positive integers. This proves the first bound in (9). This elementary bound is consistent with the classical overshoot theory [9]; we do not require a finite second moment for the tail-envelope form.

The iid strong laws give $T_k/k \rightarrow \mu_\theta$ and $\sum_{j \leq k} R_j/k \rightarrow r_\theta$. Also $\tau_k/k \rightarrow 0$ almost surely, by the tail-sum criterion and Borel–Cantelli. The residual cycle contributes at most its length, so $C_N/N \rightarrow g_\theta$ almost surely; boundedness gives convergence in mean. These are fixed-model, fixed-program assertions.

Abel summation gives the exact identity

$$(15) \quad J_{\theta, \beta}(\gamma) = (1 - \beta)^2 \sum_{N \geq 1} N \beta^{N-1} J_{\theta, N}(\gamma).$$

The coefficients sum to one. Equation (9) and the definition of c_β follow. Since $b_N \rightarrow 0$ and it is bounded, $c_\beta \rightarrow 0$. Taking suprema and infima preserves a common uniform error; it does not change the program class. Finite-prefix risks are continuous, and discounted risks are uniform limits of such risks with tail at most β^T , proving their attainment. If γ_j are asymptotically optimal for $N_j \rightarrow \infty$ or $\beta_j \uparrow 1$, their average robust risks tend to $V_{av}^\Theta(M)$. Lower semicontinuity at any convergent subnet proves the cluster-point assertion.

Finally, under (10), the right side of (14) is bounded by

$$\mathbb{E} \sum_{s=0}^N (\tau - s)_+ \leq \mathbb{E}[\tau \min\{\tau, N+1\}] \leq (N+1)^{1-p} \mathbb{E}\tau^{1+p}.$$

For the discount bound, let G have geometric distribution $\mathbb{P}(G = N) = (1 - \beta)\beta^{N-1}$. Concavity gives $\mathbb{E}(G+1)^{1-p} \leq (\mathbb{E}G + 1)^{1-p} \leq [2/(1 - \beta)]^{1-p}$. Insert the preceding overshoot bound into (15) to obtain (11). $\square$

Proposition 3.3 (Sharp time orders and the need for a tail envelope). *The orders N^{-p} and $(1 - \beta)^p$ in (11) are sharp uniformly over a bounded $1 + p$ return-moment class, even with no control decisions. A uniform first-moment bound alone does not make average reward continuous under finite-response convergence.*

Proof. A cycle has a zero-score preparation followed by either one zero-score data call, or L unit-score calls. The latter event has probability $e = L^{-1-p}$. Then

$$\mu = 2 + e(L - 1), \quad g = \frac{eL}{2 + e(L - 1)}, \quad \mathbb{E}\tau^{1+p} \leq 2^{1+p} + 2^{1+p}eL^{1+p}.$$

By call N at most N cycles can start, so $J_N \leq Ne$. Taking $L = 12N$ yields $J_N \leq g/4$, since $\mu \leq 3$, and $g \geq L^{-p}/3$. For the discount, regeneration gives

$$J_\beta = \frac{e\beta(1 - \beta^L)}{1 - (1 - e)\beta^2 - e\beta^{L+1}} \leq \frac{e}{1 - \beta}.$$

Choose an integer L between $12/(1 - \beta)$ and $13/(1 - \beta)$ when β is sufficiently close to one. Again $J_\beta \leq g/4$ and g has order $(1 - \beta)^p$. These are uniform-class lower examples, not assertions that every fixed experiment attains either worst order.

For the last assertion instead take $e = 1/L$. Finite response laws converge to the two-call zero-score cycle, while the mean lengths stay bounded by 3 and $g \rightarrow 1/3$. The limit cycle has average reward zero. The family fails uniform integrability exactly at its rare long cycles. $\square$

4. CAUSAL DEFECTS AND FINITE EXECUTABLE CERTIFICATES

We use $\text{TV}(P, Q) = \sup_B |P(B) - Q(B)|$, one half of the L^1 distance for probability densities. A cycle comparison includes its preparation, visible flags, length, declared actions, reports, performed audit marks and readouts. It is not merely a comparison of terminal hidden states.

Theorem 4.1 (Recurrence-to-defect transfer). *Suppose two completed scored-cycle laws have total-variation distance at most ε , scores in $[0, 1]$, mean lengths at least one, and each satisfies the tail bound $a(T)$ in (5). Their average risks differ by at most*

$$(16) \quad \chi_a(\varepsilon) = \min \left\{ 1, \inf_{T \in \mathbb{N}} (2T\varepsilon + 4a(T)) \right\}.$$

If both satisfy the $1 + p$ moment bound K , the bound

$$(17) \quad \chi_{p,K}(\varepsilon) = \min \{ 1, 4K^{1/(1+p)} \varepsilon^{p/(1+p)} \}$$

may be used instead. A uniform per-call discrepancy ω in the scored loss adds at most ω .

Let a parameter-independent causal simulator lift each legal target program to the source, preserving the task, actions, preparation and reset protocol, and suppose its completed scored-cycle defect is uniformly at most ε . If it stores at most S states in addition to the target's M , then

$$(18) \quad V_{\text{av,source}}^\Theta(MS) \leq V_{\text{av,target}}^\Theta(M) + \chi_a(\varepsilon) + \omega.$$

The statement is measured per target raw call. A source implementation with a different physical clock must additionally report its acquisition and time conversion; no equality of the two time averages is inferred. Serial simulator state bounds multiply. Defects on a common scored-cycle interface add under composition, before applying (16). A reverse value bound requires a reverse morphism or an independent decision-theoretic lower bound.

Proof. Put $r_T = \mathbb{E} \sum_{t \leq \tau \wedge T} \ell_t$ and $\mu_T = \mathbb{E}(\tau \wedge T)$. Each truncated functional has range $[0, T]$, so its expectation changes by at most $T\varepsilon$. The omitted reward and length each have expectation at most $a(T)$. Hence $|r - r'|$ and $|\mu - \mu'|$ are bounded by $T\varepsilon + 2a(T)$. As $0 \leq r' \leq \mu'$ and $\mu \geq 1$,

$$|r/\mu - r'/\mu'| \leq |r - r'| + |\mu - \mu'|.$$

This proves (16). Alternatively, maximally couple the two completed cycles. On a mismatch event of probability at most ε , the discrepancy of either reward or length is bounded by $\tau + \tau'$. Hölder's inequality bounds its expectation by $2K^{1/(1+p)}\varepsilon^{p/(1+p)}$, proving (17). A pointwise score discrepancy sums to at most $\omega\tau$ and is divided by μ .

For the simulator, the combined state is the pair of target and simulator states. Its report transformation, action lift, readout and reset are parameter-independent parts of one source program. Apply the cycle estimate for each fixed θ and target program, then take the model supremum and program infimum. Total variation contracts under a common kernel, so serial defects add by the triangle inequality. Neither this argument nor its resource count supplies a missing reverse simulator. $\square$

Lemma 4.2 (Primitive kernel certificates). *Consider two classical marked experiments on one common hidden space, with the same action and reset interfaces. Suppose the total variation of their one-call joint kernels for next hidden state, report, flag and performed audit is at most η , uniformly in hidden state and action. Include preparation in this bound. If the first experiment has $\mathbb{E}\tau \leq \bar{\mu}$, then the completed-cycle laws of every common controller differ by at most $\min\{1, \bar{\mu}\eta\}$.*

The same conclusion holds for finite-dimensional quantum instruments when the per-call operational instrument norm error is at most η : explicitly, one half the integrated trace-norm difference of the complete classical-quantum output is at most η for every positive input of trace one. Preparation is included. No lower bound on a report probability is needed.

If both endpoints have a common tail or moment certificate, substituting $\bar{\mu}\eta$ in Theorem 4.1 gives an all-controller average-risk modulus. A uniform stochastic-row perturbation at most η has the same conclusion. A decision approximation is charged separately by the uniform loss modulus.

Proof. For classical kernels, couple the next marked state maximally until the first discrepancy. While the paths agree, use the same controller uniforms. The probability of a first discrepancy at a call is at most η times the probability that this call is reached without a previous discrepancy. Summing over calls in the first cycle gives $\eta\mathbb{E}\tau$. Stopping flags are among the coupled reports. This is valid for discontinuous Borel controller rows, since the rows are evaluated at identical reports until failure.

For quantum instruments use unnormalized positive states, not a classical coupling of quantum states. Expand the difference of the two composed, stopped instruments by a hybrid telescoping identity. At step t , the trace mass of the first experiment still alive is $\mathbb{P}(\tau \geq t)$. The replacement error is bounded by 2η times this mass; the remaining complete instrument is trace-norm contractive on Hermitian inputs. Sum to a cutoff, include a cemetery outcome for the unfinished path, and

let the cutoff tend to infinity. The first experiment's sum of alive masses is $\mathbb{E}\tau$; almost-sure finite completion at both endpoints identifies the limiting completed-cycle laws. The half-norm convention gives the stated total variation. Report-row perturbations are classical channels acting on these outputs and obey the same telescoping bound. $\square$

The primitive comparison concerns the full marked transition, not just its marginal report density. For example, a deterministic nonlinear transition can change singularly in total variation when its parameter changes, despite continuous trajectories. In such a model this lemma does not furnish a parameter modulus. One must supply a different consistent response coupling or a certified completed-response modulus. Positivity alone does not make the effective hypotheses automatic.

Lemma 4.3 (One report-cell average). *For each a , let $\mathcal{C}_a$ be a finite Borel partition with executable membership, and discard reference-null cells. Define a reconstructed instrument by retaining the exact joint law of the report cell, next physical state, flags and audit mark, and replacing the report within its cell by an independent draw from $\sigma_a(\cdot \mid C)$. Required visible tags are never merged. Suppose its one-call operational error from the original instrument is uniformly at most κ_h .*

For every Borel γ , running γ on the reconstructed experiment has exactly the same state, action, physical-state, audit and cycle-length law as running on the original experiment the single cell table

$$(19) \quad k_{ia}^h(j \mid y) = \frac{1}{\sigma_a(C)} \int_C k_{ia}(j \mid u) d\sigma_a(u), \quad y \in C.$$

In particular, this replacement retains M and changes average risk by at most $E_h = \chi(\bar{\mu}\kappa_h)$ whenever both endpoint laws have the declared tail certificate. Here and below χ denotes either certified modulus in Theorem 4.1.

Proof. Conditional on the cell and the physical marked output, the reconstructed within-cell report is independent with the displayed law. Integration of one controller row therefore gives (19). Induction over calls, including the prescribed reset, proves the identity for every finite stopped prefix and hence for the completed cycle. This argument also proves that the reconstructed law inherits the original all-program return certificate: it equals the original law under γ^h . Lemma 4.2 and Theorem 4.1 give the error bound. The cell averages are used to prove coverage of all Borel programs; the eventual runtime algorithm classifies its actual report into a cell and uses a stored table. It does not compute conditional expectations online. $\square$

Theorem 4.4 (Same-memory finite certification). *In addition to Theorem 2.2, assume the following effective data are given with proofs or certified primitive computations. The model set is computably compact with finite s -nets $F_s \subset \Theta$. There are the executable common report partitions of Lemma 4.3, with $\kappa_h \rightarrow 0$, and finite decision ρ -nets D_ρ with uniform loss error $\omega(\rho) \rightarrow 0$. The model comparison has a controller-uniform average-risk modulus $\Omega(s) \rightarrow 0$, obtained, for instance, from Lemma 4.2. The return envelope is computable. Finite-table truncated cycle expectations are computable with rigorous intervals, uniformly over each finite list and each finite model net.*

Enumerate all denominator- Q action and transition rows on the common cells, and all D_ρ readouts. Write $J_a = |\mathcal{C}_a|$ and

$$(20) \quad \mathcal{T} \leq (Q+1)^{M|A|+M^2} \sum_a J_a |D_\rho|^M$$

for a bound on the number of tables. Suppose rigorous integration gives intervals $[l_{\gamma,\theta}, u_{\gamma,\theta}]$ for their average risks on F_s , each of width at most ζ . Define

$$L = \min_{\gamma \in \mathcal{T}} \max_{\theta \in F_s} l_{\gamma,\theta}, \quad U = \min_{\gamma \in \mathcal{T}} \max_{\theta \in F_s} u_{\gamma,\theta}.$$

Then $U - L \leq \zeta$, and

$$(21) \quad L - E_h - E_Q \leq V_{\text{av}}^\Theta(M) \leq U + \Omega(s), \quad E_Q = \chi\left(\bar{\mu} \frac{|A| + M}{Q}\right) + \omega(\rho).$$

The upper endpoint is realized by the selected finite table on the original experiment. The lower endpoint constrains every legal M -state Borel program. The interval can be made arbitrarily tight at fixed M under the stated effective data. Adding $\pm b_N$ or $\pm c_\beta$ gives corresponding finite-acquisition and discounted intervals and a same-table implementing upper bound.

Proof. Round a probability row by rounding down all but its last entry to multiples of $1/Q$ and putting the remainder in the last entry. A row of length m changes in total variation by at most m/Q . Coupling the action draw and the report-to-state draw gives per-call error at most $(|A| + M)/Q$. This may create zero entries or small new paths; all actions are legal in the continuing protocol, and no exact internal deadline is being preserved. Lemma 4.2 and the decision modulus bound the average error by E_Q . Every Borel program is thus covered by a finite table with error $E_h + E_Q$, uniformly in the model.

Let V_F be the all-Borel value on F_s and W_F the finite-table value there. Coverage and inclusion give $V_F \leq W_F \leq V_F + E_h + E_Q$. Certified intervals give $L \leq W_F \leq U$ and $U - L \leq \zeta$. Since $F_s \subset \Theta$, $L - E_h - E_Q \leq V_F \leq V_{\text{av}}^\Theta(M)$. A table attaining the finite minimum of upper endpoints has risk at most $U + \Omega(s)$ on the full family. This proves the directions in (21) without minimax exchange or a local-optimizer lower bound.

For completeness, a computable tail envelope suffices to pass from finite integration to average risk. If r_T, μ_T are the truncated expectations, then $0 \leq r - r_T \leq \mu - \mu_T \leq a(T)$ and $r_T \leq \mu_T$. Direct subtraction yields

$$|r/\mu - r_T/\mu_T| \leq a(T), \quad \mu_T \geq 1.$$

Interval arithmetic for r_T, μ_T , followed by this tail allowance, gives the required intervals. Table count (20) bounds each simplex by all its integer entries, so is intentionally crude. No polynomial planning bound is inferred. Finally use (9) for the same stationary table, not a clock-augmented implementation. $\square$

A finite table requires at most $[M|A| + M^2 \sum_a J_a] \lceil \log_2(Q + 1) \rceil$ bits for its probability entries, in addition to code and decision descriptions. Runtime scratch for random sampling, report classification and output precision is a separate resource. Offline integration and enumeration may be enormous. Finite-response domination supplies neither computable partitions nor an integration algorithm. These are explicit stronger hypotheses, verified in specific raw classes below.

5. THE GEOMETRY ACTUALLY ACQUIRED

This section fixes a known preparation and a declared exploration rule that is not altered by the compressor. It does not optimize an unrestricted control problem by comparing two filters under one policy. Let a bounded Euclidean audit variable Z_t have predictive mean $p_t = \mathbb{E}[Z_t | H_t]$ at a data call, where H_t is the full visible physical history. Independent compressor randomization does not provide extra information about Z_t conditional on H_t . Preparation calls have zero score. Set

$$(22) \quad \nu(B) = \frac{1}{\mu} \mathbb{E} \sum_{t \leq \tau, t \text{ data}} 1_{\{p_t \in B\}}, \quad B_\infty = \frac{1}{\mu} \mathbb{E} \sum_{t \leq \tau, t \text{ data}} \|Z_t - p_t\|^2.$$

The measure ν is a subprobability measure. Its mass is the actual fraction of calls spent acquiring or retaining the scored prediction. Define

$$(23) \quad q_M(\nu) = \inf_{c_1, \dots, c_M} \int \min_{j \leq M} \|x - c_j\|^2 d\nu(x).$$

Centers lie in the compact convex range allowed for decisions. A checkpoint encoder sees the full current history, but its M readouts are fixed across all calls; it is not permitted a new uncharged codebook at each time.

Theorem 5.1 (Occupation geometry and online transfer). *For the fixed exploration above, checkpoint risk satisfies*

$$(24) \quad R_{\text{cp,av}}(M) - B_\infty = q_M(\nu), \quad R_{\text{on,av}}(M) - B_\infty \geq q_M(\nu).$$

Suppose the predictive realization is a compact metric space (S, d) with initial state s_0 , Borel update U_y , and $p = f(s)$ for an L_f -Lipschitz map f . There are global M -center covers, containing s_0 , of radius r_M , with Borel nearest-center maps. Along each actual within-cycle history assume

$$(25) \quad d(U_{Y_t}(x), U_{Y_t}(x')) \leq L_t(H_t)d(x, x') \quad \text{for every } x, x' \in S.$$

All intermediate candidate states used by the algorithm must belong to S . For data ages t within a cycle put

$$G_t = \sum_{j=1}^t \prod_{u=j+1}^t L_u, \quad \mathcal{G}^2 = \frac{1}{\mu} \mathbb{E} \sum_{t \text{ data}, t \leq \tau} G_t^2.$$

If $\mathcal{G} < \infty$, one M -state recursive observer attains

$$(26) \quad R_{\text{on,av}}(M) - B_\infty \leq L_f^2 r_M^2 \mathcal{G}^2.$$

A candidate-preserving update approximation of metric error at most η and a readout approximation of Euclidean error at most ζ replace the square root on the right by $L_f(r_M + \eta)\mathcal{G} + \zeta\sqrt{\nu(S_f)}$, where $S_f = f(S)$.

If a submeasure $\xi \leq \nu$ has mass $w > 0$ and

$$(27) \quad \xi(B_r(c)) \leq Cr^d \quad \text{for all centers } c \text{ and } 0 < r \leq r_0,$$

then, whenever $(w/(2CM))^{1/d} \leq r_0$,

$$(28) \quad q_M(\nu) \geq \frac{w}{2} \left(\frac{w}{2CM} \right)^{2/d}.$$

Consequently $r_M \leq C_0 M^{-1/d}$ and a bounded $\mathcal{G}$ give matching checkpoint and online orders $M^{-2/d}$. The actual mass w remains in the constant.

All these statements also hold at a finite external acquisition budget N , using the actual measures

$$\nu_N = \frac{1}{N} \mathbb{E} \sum_{t \leq N, t \text{ data}} \delta_{p_t}$$

and the corresponding $B_N, \mathcal{G}_N$. They hold for a normalized discount with its actual occupation measure as well. Uniform rates in N or in a range of discounts require uniform versions of the displayed mass and stability certificates; they are not inferred from a terminal support alone.

Proof. Condition on the full visible history and the independent observer seed. Orthogonal projection gives

$$\mathbb{E} \|Z_t - c\|^2 = \mathbb{E} \|Z_t - p_t\|^2 + \mathbb{E} \|p_t - c\|^2.$$

For any fixed M readouts, pointwise nearest-center distance is a lower bound for every randomized encoder. A nearest-center checkpoint achieves it. Integration against the actual occupation law proves (24), including the online lower bound.

Store one cover-center label, initialized at s_0 . At a data report apply U_y to the stored center and project back to the cover. If e_t is its distance from the true predictive state, (25) gives $e_t \leq L_t e_{t-1} + r_M$, starting from zero after preparation. Hence $e_t \leq r_M G_t$. This is an estimate using a mathematical comparison state, not a register carried by the observer. Summing squares

under the actual law proves (26). With a candidate-preserving update error, replace r_M by $r_M + \eta$. Minkowski's inequality in the occupation L^2 space gives the readout allowance. No stable exact state outside the declared cover domain is silently available.

For any M centers the union of their radius- r balls has ξ -mass at most MCr^d . At $r = (w/(2CM))^{1/d}$ at least $w/2$ mass remains at distance at least r . This proves (28). The finite and discounted versions use exactly the same pointwise projection, recursion and mass proof with their own positive occupation weights. $\square$

Corollary 5.2 (Noncontractive cycles). *Suppose after a first informative update every subsequent data update is nonexpansive, the exact and approximate states coincide before that first update, and $\mathbb{E}\tau^3 < \infty$. Then the recursion in Theorem 5.1 has*

$$\mathcal{G}^2 \leq \mathbb{E}\tau^3/\mu, \quad \sup_{N \geq 1} \mathcal{G}_N^2 \leq 2\mathbb{E}\tau^3,$$

and the same second bound for every normalized discount. Strict per-step contraction is unnecessary. The cycle may have a polynomial tail.

Proof. At the first informative report the error is at most the cover radius, regardless of a larger Lipschitz constant for that first update, since the prior error is zero. Subsequent errors are at most their data age times the radius. The sum of squared ages in one cycle is at most τ^3 . For a finite horizon, sum over renewal epochs $s \leq N$. Each integer is an epoch with probability at most one, and its subsequent cycle is fresh. Thus the expected sum divided by N is at most $(N+1)\mathbb{E}\tau^3/N$. The discounted bound follows by the positive Abel mixture of finite-horizon bounds. $\square$

The global covering, the acquired small-ball mass, and the stability condition play different roles. Report-kernel continuity is not a substitute for any of them: it is used for model and finite-precision transfer in Section 4. Both raw geometric verifications below prove this continuity independently. Under their effective certificates, a finite table can approximate the best Borel risk to $o(M^{-2/d})$ at the same M ; this yields a finite executable upper without identifying $\log_2 M$ with its whole program description.

Theorem 5.3 (Renewal holding geometry). *Prepare a mark V in a compact set $K \subset \mathbb{R}^m$ with law π , and an integer $L \geq 1$ with any conditional law given V , $\mathbb{E}L < \infty$. Charge one zero-score preparation call. The first of L data calls reports V with a visible load tag; the remaining reports are blank holding tags. The audit target is V at every data call, and the observer is overwritten at the next preparation. Then, at average risk,*

$$(29) \quad R_{\text{cp}}(M) = R_{\text{on}}(M) = q_M(\nu), \quad \nu(dv) = \frac{\mathbb{E}[L \mid V = v]}{\mathbb{E}(L + 1)} \pi(dv).$$

At finite N and at any normalized discount the identical equality holds with the corresponding actual holding-occupation measure. For an arbitrary family of such preparations, including nondominated families, the robust value is the infimum over one common M -center set of the supremum of these integrals.

Proof. Every output of every M -state observer belongs to its one set of M readouts. Pointwise nearest-center distance lower-bounds its loss at each data call, including observers that try to count how long the mark has survived. At the load tag choose a nearest-center label and then retain it on every holding tag. This uses M labels, not M plus a clock: the load and hold events are physically different visible reports. The same encoder works at every parameter for fixed centers. Integration proves each equality. Compactness of K permits centers to be chosen in its compact convex hull. This proof is direct and does not infer compactness of general nondominated adaptive controllers from Lemma 3.1. $\square$

For example, let π be uniform on $[0, 1]$ and $L = \lceil V^{-\alpha} \rceil$, $0 < \alpha < 1$, with an arbitrary value at the null point $V = 0$. The occupation mass near zero has order $r^{1-\alpha}$, although preparation mass has order r . On an interval bounded away from zero the occupied density is bounded and positive. The small-ball lower there and the global interval cover give $q_M(\nu) \asymp M^{-2}$. Also $\mathbb{E}\tau^{1+p} < \infty$ whenever $\alpha(1+p) < 1$. Thus this nonuniform, possibly infinite-third-moment class has an exact acquired-memory law without accumulating recursive error. A compact d -regular Cantor preparation with $1 \leq L \leq L_0$ instead gives the noninteger order $M^{-2/d}$ by the same actual-mass proof. Unbounded length bias must be analyzed from (29), not by reusing the unweighted preparation dimension.

6. INDEPENDENT RAW-KERNEL VERIFICATIONS

6.1. Measurable densities and hidden-state experiments.

Lemma 6.1 (A measurable kernel density). *Let U, Y be standard Borel spaces, σ a probability measure on Y , and $u \mapsto K_u$ a measurable finite kernel with $K_u \ll \sigma$ for every u . There is a jointly measurable density $f(u, y)$ representing every K_u up to its own σ -null set. Consequently, sequential mixtures of such kernels have jointly measurable finite-word densities, even when hidden transitions depend on earlier reports.*

Proof. Choose increasing finite partitions $\mathcal{P}_n$ generating the Borel sigma-field of Y . On a positive-mass cell C set $f_n(u, y) = K_u(C)/\sigma(C)$ and set it to zero on null cells. Each f_n is jointly measurable. For fixed u , if f_u is any Radon–Nikodym density, $f_n(u, \cdot)$ is its conditional expectation on $\mathcal{P}_n$. Martingale convergence gives $f_n \rightarrow f_u$ almost surely and in $L^1(\sigma)$. The liminf, set to zero where infinite, is the required joint version. The exceptional set may depend on u ; joint measurability, not a common exceptional set for all u , is the assertion. Apply standard Borel disintegration to the hidden-history kernel at each step and integrate this joint density against the conditional hidden law. Iterating conditional kernels gives the product of the successive report densities. One does not condition on a hidden path as though it were independent of the reports. These version and disintegration facts agree with the standard treatment in [6, 2]. $\square$

Theorem 6.2 (Hidden-state realization). *Take a standard Borel hidden state, a prepared law, and marked causal transition kernels producing report, audit and boundary flag. Suppose that at each fixed action the conditional report kernel, given the hidden state and visible history, is dominated by a common σ_a . Suppose the charged joint preparation/observer-reset protocol and the all-controller return envelope (5) hold. Then this experiment verifies Theorem 2.2, without a finite-dimensional belief assumption or a filter-contraction assumption.*

A complete effective verification is available for the following subclass. Apart from an independent renewal clock, the hidden space and audit alphabet are finite. The continuous part of a report is a finite union of compact cubes, with a common Lebesgue reference and finitely many visible atoms. Joint marked transition densities have uniform computable bounds on their first report derivatives and a uniform computable parameter modulus on a computably compact parameter set. The clock law and its tail envelope have certified truncations, and the preparation, density entries and compact readout loss have certified integrations. Then Theorem 4.4 applies, with report reconstruction error $\kappa_h \leq Ch$ on a mesh of diameter h and a primitive model modulus obtained by integrating the stated entrywise parameter bounds.

Proof. Lemma 6.1 applied to the sequential conditional mixtures gives (2) for every finite intervention word. The performed payoff and survival submeasures are dominated by that report law. The renewal protocol supplies iid cycles conditional on the fixed parameter, and the assumed envelope supplies precisely (5). This proves the exact verification without invoking a realization-specific risk theorem.

For the effective subclass, keep each atom and each visible flag in its own cell. On a cube cell, replace each matrix entry density $k(y)$ by its average. A uniform derivative bound B gives $|k(y) - k_C| \leq Bh$. Sum over the finitely many next-state, audit and flag indices and integrate over the reference volume. Half of the resulting L^1 bound is the required operational $\kappa_h \leq Ch$. The identical entrywise integration gives the parameter error $\eta(s)$, and Lemma 4.2 gives

$$(30) \quad E_h \leq \chi(\bar{\mu}Ch), \quad \Omega(s) \leq \chi(\bar{\mu}\eta(s)) + \omega_{\text{loss}}(s).$$

The clock is retained exactly by the report reconstruction. It belongs to the physical experiment, not to an uncharged observer counter. For a finite integration horizon its possible termination times below that horizon form a finite list, with the surviving tail retained as a single marked case. Finite products of computable density entries and finite probability tables have certified integrals on the cells. The uniform clock tail then bounds the omitted cycle reward and length, exactly as in the proof of Theorem 4.4. This derives all of that theorem's effective inputs.

For Gaussian reports, the same construction works when the parameter family gives uniform derivative bounds on each truncation cube and a computable uniform tail bound. Keep the entire tail as a genuine cell, rather than condition it away. A coarse tail reconstruction has operational error at most its tail mass, while compact-cell errors are bounded as above. Certified Gaussian integrals and a decreasing tail bound produce $\kappa_h \rightarrow 0$. A parameter change in a hidden deterministic transition is not automatically small in operational norm; that comparison must be verified independently. $\square$

Theorem 6.3 (A noncontractive hidden predictive geometry). *Let the hidden state have $d + 1$ values, $d \geq 1$, and a known preparation $\pi_i > 0$. The first data report has reference uniform measure on $[-1, 1]^d$ and likelihoods*

$$(31) \quad g_i(y) = 1 + \eta y_i \quad (1 \leq i \leq d), \quad g_0(y) = 1 - \eta \sum_{i=1}^d y_i, \quad 0 < \eta < 1/(2d).$$

After this report, each next hidden transition is a known stochastic matrix Q_a , selected by a publicly reported innovation independent of the hidden state, under a fixed exploration. The audit target is the current hidden-state indicator vector. The cycle contains $L \geq 1$ data calls, with L independent of the hidden experiment and $\mathbb{E}(L + 1)^3 < \infty$. Then the actual checkpoint and online squared-prediction excess risks have matching order

$$(32) \quad M^{-2/d}$$

at average risk, uniformly at every external $N \geq 2$, and uniformly for $\beta \geq 1/2$. All observer states, including the reset state, are counted. The matrices may include permutations and need not contract.

Proof. All likelihoods are strictly positive, integrate to one, and have constant sum $d + 1$. The first posterior is $p_i = \pi_i g_i / (\sum_j \pi_j g_j)$. It has the explicit inverse

$$g_i = \frac{(d + 1)p_i / \pi_i}{\sum_j p_j / \pi_j}, \quad y_i = (g_i - 1) / \eta \quad (1 \leq i \leq d).$$

On the compact likelihood cube this is a smooth bijection with nonsingular derivative between its d -dimensional interiors; the derivative and inverse derivative are uniformly bounded. The report density $\sum_i \pi_i g_i$ is bounded above and below. The first posterior law therefore has a bounded density on the simplex's affine span, and positive mass. Its conditional future-test geometry has dimension d : the indicator audit probabilities separate posterior states. No covariance or Fisher geometry has been postulated.

Subsequent updates are $p \mapsto pQ_a$, which are nonexpansive in ℓ^1 . The entire simplex has global covers of radius $CM^{-1/d}$ including the initial prior, and the readout into Euclidean indicator

means is Lipschitz. The first likelihood update is globally Lipschitz because its denominator is uniformly positive; its initial error is zero. Corollary 5.2 therefore supplies a uniform occupation stability bound. The first report law has total-variation continuity in its prior by integrating $\sum_i (p_i - p'_i) g_i$; later public reports are independent of that prior. This independently verifies report continuity.

There is one first data report per cycle. At average risk its acquired submeasure is the first posterior law divided by $\mu = \mathbb{E}(L + 1)$. For finite N , renewal epochs $s \leq N - 2$ each supply this same law. Their expected number exceeds $(N - 2)/\mu$ by the stopped iid sum identity; it is also at least one. Thus its weight divided by N is at least $1/(2\mu)$ for every $N \geq 2$. At discount β its weight is $\beta(1 - \beta)/(1 - \mathbb{E}\beta^\tau) \geq \beta/\mu$. For $\beta \geq 1/2$ the same lower bound follows. Apply the actual small-ball inequality and the uniform global-cover recursion of Theorem 5.1. Small M are absorbed into fixed positive constants. Fixed computable coefficients and a clock with certified tails also verify the effective subclass above, yielding finite executable tables with the same risk order and unchanged M . $\square$

6.2. Noncommuting quantum instruments. A finite-dimensional quantum system below may be driven by a separate classical renewal clock. The clock's storage and physical cost belong to the specified experiment. They are not counted as observer memory, nor treated as costless to a simulator that must reproduce them.

Theorem 6.4 (Quantum realization and operator certificates). *Let a finite-dimensional quantum system be prepared by a density matrix and evolve under marked completely positive instrument densities $\mathcal{I}_{\theta,a}(y, z, b)$, whose sum and integral are trace preserving. The visible report contains y, b ; the actual audit mark z is not feedback. All densities are measurable with respect to common σ_a , and the joint reset protocol has a controller-uniform return envelope. Then Theorem 2.2 applies to one classical M -state observer, including projective zeros, pure preparations and noncommuting actions.*

For a complete effective realization, it suffices that the Kraus matrices and preparation matrices have computable uniform approximation bounds on a finite union of report cubes and atoms, with a certified renewal-clock tail and certified finite integrals. If $A_j(y)$ and $B_j(y)$ are corresponding Kraus approximations, the operational one-call error is bounded by one half of

$$(33) \quad \sum_j \int (\|A_j(y)\| + \|B_j(y)\|) \|A_j(y) - B_j(y)\| d\sigma(y).$$

The same formula applies to parameter approximations. Report-cell averaging of the full instrument gives $\kappa_h = O(h)$ under uniform report Lipschitz bounds, and these estimates give the E_h, Ω and certified finite integrals in Theorem 4.4 through Lemma 4.2.

Proof. For each finite intervention word, compose the unnormalized completely positive densities and take their trace, summing unobserved audit marks. This gives a positive L^1 report density of total mass at most one for a stopped word. Performing the actual audit and weighting its outcomes by $\ell(d, z)$ gives a dominated payoff density. No normalization by a possibly zero report probability occurs. Preparation and reset give independent cycles conditional on the same fixed model, so the central hypotheses are verified.

For a positive input ρ of trace one,

$$\|A\rho A^* - B\rho B^*\|_1 \leq (\|A\| + \|B\|)\|A - B\|,$$

by adding and subtracting $B\rho A^*$ and using the ideal property of the trace norm. Summation and integration give (33), with the classical mark retained. Cell-averaging the complete instrument, rather than averaging and then squaring its Kraus matrices, preserves complete positivity and

normalization. An instrument Lipschitz bound integrates to an $O(h)$ operational error. Each finite-table response is a finite sum of traces of products of computable matrices; certified quadrature and the renewal tail give the finite risk intervals. This is an explicit operator-to-certificate implication. It does not assert that every measurable Kraus density is computable. The argument is finite-dimensional; no unbounded-operator closure is used. $\square$

Theorem 6.5 (Noncommuting, changing-rank acquired geometry). *On a q -level system, $q \geq 2$, let $d = q^2 - 1$ and let $H_1, \dots, H_d$ be a basis of traceless Hermitian matrices. Choose a symmetric cube $C \subset \mathbb{R}^d$ small enough that*

$$E(y) = I + \sum_{j=1}^d y_j H_j$$

is uniformly positive. Prepare I/q . With probability $1 - \omega > 0$, the first data instrument has report $y \in C$, uniform reference, and Kraus density $\sqrt{E(y)}$. With probability $\omega \in [0, 1)$, it instead performs a tagged rank-one projective measurement $\{P_j\}_{j=1}^q$. These two tagged instruments are mixed with their stated weights.

At each data call a fixed informationally complete POVM $\{F_z\}$ is actually performed for the audit and its outcome is not feedback. Its average back-action is $\mathcal{A}(X) = \sum_z \sqrt{F_z} X \sqrt{F_z}$. Subsequent visible innovations select known unitaries independently of the system, so the pre-audit predictive state updates by $X \mapsto U\mathcal{A}(X)U^$. Noncommuting unitaries are allowed. Let the independent data-call lifetime $L \geq 1$ satisfy $\mathbb{E}(L+1)^3 < \infty$.*

For predicting the audit indicator vector under squared loss, checkpoint and online excess risks have matching order $M^{-2/(q^2-1)}$ at average risk, uniformly in $N \geq 2$, and uniformly in $\beta \geq 1/2$. The proof uses actual continuous mass $(1 - \omega)$ and does not replace the rank-one component by a smooth density. The conclusion is for this fixed exploration, not for unrestricted quantum control.

Proof. The continuous first instrument is normalized because $\int E(y) d\sigma(y) = I$. Under I/q the report law is exactly uniform, since $\text{tr } E(y)/q = 1$, and its pre-audit posterior is $X = E(y)/q$. Thus the acquired first-state law is affine full-dimensional in the d -dimensional trace-one space, with bounded density on its compact image. The projective component yields the pure states P_j with their actual probabilities. The continuous first-state submeasure has mass $1 - \omega$, not one.

For an explicit informationally complete audit, take $F_{j,\pm} = (I \pm cH_j)/(2d)$, with $c > 0$ small enough for positivity. They sum to I . The map $f(X) = (\text{tr } F_z X)_z$ is injective on trace-one matrices, with upper and lower Lipschitz constants on their difference space. Consequently these actually executable audit probabilities determine the predictive state, and the first continuous response law has dimension d . The following back-action is included in the raw marked instrument; incompatible audits are not sampled jointly.

Trace-preserving positive maps are contractive in trace norm on Hermitian differences, as follows by applying positivity and trace preservation to the positive and negative parts. Hence $X \mapsto U\mathcal{A}(X)U^*$ is nonexpansive. The whole density-matrix space has global trace-norm covers of radius $CM^{-1/d}$ containing I/q , and f is Lipschitz. The first report map is regular on its continuous positive branch and defined at the projective reports; only the exact prior is used before this first update. Corollary 5.2 gives the required stability. First report probabilities are linear in the input matrix, giving total-variation continuity; later public reports are independent of it.

The continuous first report occurs with occupation weight $(1 - \omega)/\mu$ at average risk, and at least $(1 - \omega)/(2\mu)$ in the finite and discounted ranges proved in Theorem 6.3. Push this actual submeasure through f and apply Theorem 5.1. Its mass enters the lower constant as in (28). The atom and subsequent mixed-state portions remain in the upper cover and in the actual risk. The ambient trace-one space was derived here from the raw instrument and separating audit, not

supplied as an unexplained predictive geometry. At $\omega = 1$ the full-dimensional lower does not follow and is not asserted. $\square$

Corollary 6.6 (Singular references and return laws). *The exact hidden and quantum verifications remain valid for a common Cantor report reference, a countable report law, or finitely many observed rank strata. The effective verifications remain valid when executable partitions have certified omitted mass and instrument oscillation on each retained cell. A common renewal clock with $\mathbb{P}(L \geq n) \leq Cn^{-r}$, $r > 1+p$, supplies a finite $1+p$ moment; $r > 3$ supplies the noncontractive geometric criterion. All tail, continuous and stratum probabilities retain their original mass.*

Proof. The measurable-density and tensor arguments use product probability measures and L^1 approximation, not Lebesgue density. For a discarded tail cell of mass at most u uniformly in physical inputs, replacement costs at most u in operational total variation; keep that cell as an actual report rather than condition on its complement. On the retained cells use their certified oscillation. For the lifetime moment, the identity expressing $\mathbb{E}L^s$ as a sum of successive power increments times $\mathbb{P}(L \geq n)$ converges for $r > s$. Observed rank tags are part of the common reference, and the completely positive instrument estimate does not require invertible effects. Mutually singular translated Cantor families need not have one common reference, and are not covered merely by this corollary. $\square$

7. A SHARP JOINT RESOURCE LAW

The defect modulus concerns an attained experiment difference, not an arbitrary permissible error tolerance. The following single task realizes it together with a geometric memory coordinate and an inaccessible calibration coordinate.

Proposition 7.1 (Memory, calibration and a long-cycle defect). *Let π be a probability law on a compact subset $K \subset \mathbb{R}^m$, and let $\theta \in \{0, 1\}$ and $u \in \{-\delta, \delta\}$ be fixed inaccessible parameters, $0 \leq \delta \leq 1$. At each cycle prepare $V \sim \pi$, independently of the parameters and clock. Charge one zero-score preparation. With probability $1 - e$ there is one data call; with probability e there are $L \geq 1$ data calls. The type is visibly tagged, all later data reports are blank, and the observer is overwritten at the next preparation. The target first report is (V, θ) in either type. The source first report is (V, θ) in the short type and $(V, \perp)$ in the long type. At every data call the actual audit is (V, θ, u) , not feedback. Its loss is*

$$\|d_V - V\|^2 + (d_b - \theta)^2 + (d_u - u)^2.$$

Let $\mu = 2 + e(L - 1)$ and $a = (\mu - 1)/\mu$. For every $M \geq 3$, with $J = \lfloor M/3 \rfloor$, the source's optimal robust average risk obeys

$$(34) \quad aq_M(\pi) + \frac{eL}{4\mu} + a\delta^2 \leq \mathcal{R}_{\text{av}}(M, e, L, \delta) \leq aq_J(\pi) + \frac{eL}{4\mu} + a\delta^2.$$

The upper is implemented by one common finite-state encoder whenever the chosen J -center classifier and centers have the declared executable precision; otherwise it is the exact Borel upper and is finitely approximable for computable regular or Cantor codebooks.

Under task-preserving, cycle-aligned causal simulation, with the target first report required before any later acquisition, the source-to-target completed-cycle deficiency is exactly $\varepsilon = e/2$. The simulator may not use audit marks as feedback. For $0 < p \leq 1$, choose $e = 2\varepsilon$ and $L = \lfloor e^{-1/(1+p)} \rfloor$. If $q_M(\pi) \asymp M^{-2/d}$, then for sufficiently small $\varepsilon > 0$,

$$(35) \quad \mathcal{R}_{\text{av}}(M, \delta, \varepsilon) \asymp M^{-2/d} + \delta^2 + \varepsilon^{p/(1+p)}, \quad M \geq 3,$$

with a uniform $1 + p$ cycle moment. The constants depend on the declared geometric law and p , not on M, δ, ε . In particular the exponent in (17) is sharp in its moment class.

For the same stationary observer class at finite real acquisition budget N , Theorem 2.2 gives

$$(36) \quad |\mathcal{R}_N(M, \delta, \varepsilon) - \mathcal{R}_{\text{av}}(M, \delta, \varepsilon)| \leq 2HKN^{-p},$$

where $H = \text{diam}(K)^2 + 5$ bounds the loss for decisions in $\text{conv}(K) \times [0, 1] \times [-\delta, \delta]$. In particular the calibration loss is at most $4\delta^2$, not δ^2 for an arbitrary decision. An analogous bound holds for vanishing discount. The term in (36) is a signed finite-acquisition comparison, not an asserted positive learning penalty.

Proof. Any M -state observer has at most M vector readouts, so its vector loss is at least the nearest-center loss. The clock is independent of V , giving the occupied vector law $a\pi$. Average the fixed parameters $\theta = 0, 1$ and $u = \pm\delta$ only to prove a minimax lower. This averaging does not change them during a path. On a long source cycle, no visible report contains information about θ , and no information survives from another cycle; a fair bit therefore contributes at least $1/4$ per data call. The unobserved calibration sign contributes at least δ^2 at every data call. Squared coordinates add, proving the lower in (34) for every randomized program.

For the upper, use J vector centers and bit readouts $0, 1, 1/2$, with calibration output zero. A load report selects a center and either the observed bit or the erasure readout. Holding reports retain that label. There are $3J \leq M$ labels; the distinguished reset label is one of them and has zero-score preparation, so no extra phase state is needed. The vector loss is $aq_J(\pi)$, the bit loss is exactly $eL/(4\mu)$ for each fixed θ , and the calibration loss is exactly $a\delta^2$ for each fixed u . Near-optimal centers suffice if the infimum notation is used; compactness also gives exact centers.

On a source erasure, a causal simulator can issue an independent fair bit, and otherwise issue the reported bit. Everything else is copied with its actual cycle type, length and audit law. For either fixed θ , the probability that its first bit is wrong is $e/2$. Hence the full-cycle total variation is at most $e/2$. Conversely, conditioned on the erasure type, the simulator's first-bit distribution is common to both values of θ . At least one parameter has an incorrect-bit probability at least $1/2$. The type has probability e and the target bit is certain, giving deficiency at least $e/2$. The task-preserving requirement prevents rewriting the later true audit. The first-report timing prevents using a future acquisition to infer the bit. No assertion is made for a different cross-cycle, nonreset simulation interface.

For e small, L is between one half and one times $e^{-1/(1+p)}$, so $eL \asymp e^{p/(1+p)}$ and $2 \leq \mu \leq 3$. Also

$$\mathbb{E}\tau^{1+p} = (1-e)2^{1+p} + e(L+1)^{1+p} \leq 2^{2+p}.$$

Thus a stays between $1/2$ and $2/3$, $q_J(\pi) \asymp q_M(\pi)$ under the declared power law, and (35) follows. Setting the vector and calibration tasks to zero leaves an exact loss $eL/(4\mu)$ even against unrestricted within-cycle history. This supplies the defect lower order independently of quantization. Finally rescale the bounded summed loss by H , apply the uniform renewal theorem and undo the scaling to prove (36). $\square$

The lifetime clock in this example is a physical experimental resource. It is not a counter secretly stored by the observer: the observer merely follows visible load, hold and reset reports. Simulating the clock requires its own stated resources. Similarly, the audit is an actual marked kernel operation in this formulation; implementing it as a separate physical call changes the acquisition ledger and must be charged accordingly. The example proves a selected matched slice of the resource region. It does not prove joint optimality of program description, workspace, physical time and minimal simulator state.

8. COMPARISON AND SCOPE

Witsenhausen's static reduction [12] provides an independent reference representation for finite causal experiments. Strategic-measure compactness and information-structure methods [16, 14]

establish powerful existence and optimality principles for larger policy spaces. They do not eliminate the obligation to check that a restriction to one fixed number of *reused* report rows is closed and that its finite responses are continuous. Lemma 3.1 states that check. Remark 3.2 exhibits the failure without the product-history assumption. Once that continuity and a uniform cycle tail have been proved, the attainment and finite-model intersection steps are standard compact analysis, not a new abstract separation principle.

Average-cost partially observed control already has a substantial theory. Yu and Bertsekas [13] study near optimality of the set of finite-state controllers for average-cost POMDPs. Their approximation over finite-state controllers is not a fixed- M robust optimum with a same- M certificate. Yüksel’s direct history-space approach [15] supplies discounted and average-cost existence conditions without first reducing to a belief state, together with finite-window approximation results. Demirci, Kara and Yüksel [3] derive average-cost optimality through contraction properties of nonlinear filters. These approaches concern different hypotheses and policy classes; a compulsory observer reset is neither a consequence of filter contraction nor a replacement for their unrestricted-policy conclusions. Our uniform passage in Theorem 2.2 is instead for a single common stationary finite alphabet, under a return certificate of the physical-plus-observer protocol. Theorem 5.1 then relates its finite-resource predictions to actual cycle occupation.

The finite-memory approximation results of Kara and Yüksel [7] connect finite windows to filter stability. Kim [8] gives a policy-conditional Wasserstein comparison under the same closed-loop execution; such a result must not be described as a comparison of independently optimized control policies. This distinction is respected here: Theorem 2.2 optimizes its full declared common-program class, whereas the geometric prediction Theorem 5.1 explicitly fixes exploration. Hudák and coauthors [4] extract robust finite-state controllers from learned recurrent policies for hidden-model POMDPs. Their computationally evaluated controllers and the inductive synthesis of Andriushchenko and coauthors [5] address practical synthesis, not the all-Borel coverage lower of Theorem 4.4. Conversely, our exhaustive bound does not supply their practical scalability. Small-policy complexity [10] is a reason to keep finite certifiability separate from efficient synthesis.

The exact renewal theorem admits arbitrary standard Borel hidden dynamics and arbitrary fixed-model families under finite-history product domination and a common return envelope. The effective theorem additionally needs computable primitive or response certificates; the displayed hidden and quantum implications supply them for the stated subclasses. The geometric theorem needs actual mass, a global cover, and candidate-valid occupation stability. The direct holding theorem covers nondominated preparation families but does not remove domination from general adaptive control. These are distinct levels of assumptions, not interchangeable uses of the word “general.”

The remaining mathematical problems include an unrestricted nonreset average-risk theorem under comparably primitive hypotheses; a broad nondominated adaptive common-controller theory; a near-necessary intrinsic criterion replacing global covers and sufficient stability; efficient synthesis; and a complete matched region for program length, scratch, precision, planning time, physical time and minimal simulator state. No theorem here asserts a large-deviation principle from total-variation proximity, an infinite-path law from finite domination, or an unbounded-operator closure from finite matrices. The earlier discounted theorem [11] and earlier exact-deadline and causal-allocation results retain their own protocols. They are preserved as complete technical companions, not used as unproved premises of the new average-risk theorem.

REFERENCES

- [1] S. Asmussen, *Applied Probability and Queues*, second ed., Springer, 2003.
- [2] D. P. Bertsekas and S. E. Shreve, *Stochastic Optimal Control: The Discrete-Time Case*, Academic Press, 1978.

- [3] Y. E. Demirci, A. D. Kara and S. Yüksel, Average cost optimality of partially observed MDPs: contraction of nonlinear filters and existence of optimal solutions and approximations, *SIAM J. Control Optim.* **62** (2024), 2859–2883. DOI: 10.1137/24M1643736.
- [4] D. Hudák, M. F. L. Galesloot, M. Tappler, M. Kurečka, N. Jansen and M. Češka, Finite-state controllers for (hidden-model) POMDPs using deep reinforcement learning, arXiv:2602.08734, 2026.
- [5] R. Andriushchenko, M. Češka, S. Junges and J.-P. Katoen, Inductive synthesis of finite-state controllers for POMDPs, *Proc. Mach. Learn. Res.* **180** (2022), 85–95.
- [6] O. Kallenberg, *Foundations of Modern Probability*, third ed., Springer, 2021.
- [7] A. D. Kara and S. Yüksel, Near optimality of finite memory feedback policies in partially observed Markov decision processes, *J. Mach. Learn. Res.* **23** (2022), no. 11, 1–46.
- [8] M. Kim, Finite memory belief approximation for optimal control in partially observable Markov decision processes, arXiv:2601.03132, 2026.
- [9] G. Lorden, On excess over the boundary, *Ann. Math. Statist.* **41** (1970), 520–527. DOI: 10.1214/aoms/1177697092.
- [10] M. Mundhenk, The complexity of optimal small policies, *Math. Oper. Res.* **25** (2000), 118–129. DOI: 10.1287/moor.25.1.118.15214.
- [11] Q. Qi, *General Theta Foundations I: Acquired Geometry and Causal Resource Transfer—Discounted Raw-Response Theorem*, complete technical companion B, 2026.
- [12] H. S. Witsenhausen, Equivalent stochastic control problems, *Math. Control Signals Systems* **1** (1988), 3–11. DOI: 10.1007/BF02551232.
- [13] H. Yu and D. P. Bertsekas, On near optimality of the set of finite-state controllers for average cost POMDP, *Math. Oper. Res.* **33** (2008), 1–11. DOI: 10.1287/moor.1070.0279.
- [14] H. Yu, On strategic measures and optimality properties in discrete-time stochastic control with universally measurable policies, *Math. Oper. Res.* **49** (2024), 1734–1760. DOI: 10.1287/moor.2022.0188.
- [15] S. Yüksel, Another look at partially observed optimal stochastic control: existence, ergodicity, and approximations without belief-reduction, arXiv:2301.11244, version 3, 2024.
- [16] S. Yüksel and N. Saldi, Convex analysis in decentralized stochastic control, strategic measures, and optimal solutions, *SIAM J. Control Optim.* **55** (2017), 1–28. DOI: 10.1137/15M1049129.